

DESCRIPTIVE TEST.

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## 1. SMART HOME AUTOMATION SYSTEM:

### INTRODUCTION:

Smart Home automation system controls lights, temperature, security based on the user's behavior, and the environmental conditions. The intelligent agents receives the information from the motion sensors temperature sensors etc.

Each intelligent agent is used for different types of tasks.

### 1) SIMPLE - REFLEX AGENT:

Simple - Reflex agent is used to control the current



input. It has no memory of the past.

- \* Eg: i) When motion is detected  $\rightarrow$  Turn on light.
- ii) When temperature  $> 30^{\circ}\text{C}$  then, task performed accordingly.

\* Advantages: Easy to implement, low computational cost.

\* Disadvantages: No past memory, Cannot connect and understand the user's preference, may make poor decisions in environment conditions.

## 2. MODEL - BASED AGENT:

Model - Based Agent maintains the internal memory (model). This Agent knows the past and also predicts the future.

\* Advantages: Better future predictions, Good decision making.

\* Disadvantages: Requires more memory, Difficult to implement.



### 3. GOAL - BASED AGENT:

Goal - Based Agents helps to <sup>selects</sup> ~~make~~ or do a actions based on the specific tasks or assignment.

\* Advantages: It makes better decisions, even when no information is available it works

\* Disadvantages: When multiple factors occur it will not be able to decide and perform.

### 4. UTILITY - BASED AGENT:

Utility based agent will choose a overall best performance out of a multiple factors.

\* Advantages: Can be used or applied in large complex tasks, easy decision making.

\* Disadvantage: Difficult / complex to implement, more memory usage, high cost.



| Factors                                       | Simple reflex     | Model-Based       | Goal Based               | Utility Based                 |
|-----------------------------------------------|-------------------|-------------------|--------------------------|-------------------------------|
| Memory                                        | No                | Yes               | Yes                      | Yes.                          |
| Leaves                                        | No                | Limited           | No                       | Yes (with learning)           |
| When compared to smart home automation system | Better automation | Better monitoring | Planning & Optimization. | Best for overall performance. |

**MOST EFFECTIVE AGENT FOR THIS SYSTEM:**

"Hybrid Intelligent Agent" is the best choice out of all because all the other four perform some specific work like immediate response, planning, memory, searching etc.. So combining all this will make it more better and efficient.



## UNIFORM COST SEARCH (UCS):

- \* Uniform Cost Search algorithm gives the lowest cumulative cost.

- \* It guarantees the low cost path.

### Advantages:

- \* Finds the minimal cost.

- \* Predicts the optimal path.

### Disadvantages:

- \* Repeats each node again and again by backtracking to find the shortest path.

- \* Requires large amount of memory.

### WORKING:

- \* Starts with the initial node.

- \* Inserts the start node to the priority queue.

- \* Adds the next node according to the cost.

- \* Repeats the process.

- \* Continue increasing the limit.



### JUSTIFICATION:

A hybrid approach should contain

- \* Comfort
- \* Energy efficiency
- \* Security
- \* Adaptability.

### CONCLUSION:

Therefore hybrid Intelligent Agent is the best choice as it give comfort, energy efficient, security performance and also adaptability.

2.

### MAZE:

#### INTRODUCTION:

A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. So this can be performed by using search algorithms to solve this problem.



\* Stop when the exit is found.

### COMPLEXITY.

\* Time Complexity :  $O(b^{1 + \lceil C^*/\epsilon \rceil})$

\* Space Complexity :  $O(b^{1 + \lceil C^*/\epsilon \rceil})$

where :

$b \Rightarrow$  branching factor.

$C^* \Rightarrow$  optimal path cost.

$\epsilon \Rightarrow$  minimum step cost.

### ITERATIVE DEEPENING SEARCH (IDS):

IDS repeatedly performs Depth-limited search while gradually increasing the depth limit until the goal is found.

### Advantages :

\* Very low memory usage.

\* Complete for finite search spaces.

### Disadvantages :

\* Repeats node expansion.

\* Not optimal when path costs differ.



COMPLEXITY:

\* Time Complexity:  $O(b^d)$

\* Space Complexity:  $O(bd)$ .

where:

$b \Rightarrow$  branching factor.

$d \Rightarrow$  depth of the shallowest solution.

MOST EFFECTIVE AGENT:

"Goal - Based Agent"

because:

\* It focuses on reaching the exit.

\* It can use search algorithms efficiently.

\* It continuously updates its plan as new information is discovered.

MOST EFFECTIVE SEARCH:

UNIFORM COST SEARCH (UCS).

because:

\* Finds the least cost path.

\* Suitable when movement costs vary.



\* Guarantees an optimal solution.

### CONCLUSION:

A model - Based Goal - Based Intelligent Agent using Uniform Cost Search (UCS) is the most effective solution for navigating an unknown maze. It can learn the environment, remember explored paths etc.

